

Assignment 5B

ELO Calculations

Approach

The idea is to reuse the parsing from Project 1 to extract a player's pre-rating and the list of opponents and apply the ELO formula to find what the score of each player should be depending on the level of his/her opponents and to compare it with his/her actual score. Sorting the differences of the two scores gives the top five over- and under-performers.

Just a couple of things to note: I am calculating each player's pre-tournament rating for every round rather than updating it after each round since this is all the information I have; so this is a simplification of how ELO system operates. Also, I should take out byes and unplayed rounds from the calculations, which is the same problem as in the previous project. Finally, since there are a number of variants of the Elo formula, I should cite the exact formula I am using.

Code Base

```
library(dplyr)
```

```
Attaching package: 'dplyr'
```

```
The following objects are masked from 'package:stats':
```

```
filter, lag
```

```
The following objects are masked from 'package:base':
```

```
intersect, setdiff, setequal, union
```

```
library(stringr)
```

```
library(knitr)
```

```
players_df <- read.csv("https://raw.githubusercontent.com/daanishrasheed/DATA  
607/refs/heads/main/Assignment%205B/tournament_ratings.csv")
```

```
kable(head(players_df))
```

pair	name	state	points	pre_rating	avg_opponent_pre_rating
1	GARY HUA	ON	6.0	1794	1605
2	DAKSHESH DARURI	MI	6.0	1553	1469
3	ADITYA BAJAJ	MI	6.0	1384	1564
4	PATRICK H SCHILLING	MI	5.5	1716	1574
5	HANSHI ZUO	MI	5.5	1655	1501
6	HANSEN SONG	OH	5.0	1686	1519

The Elo rating system calculates the probability of winning a match against a particular opponent solely using the ratings difference of the two players. Here the equation used is:

$$E_A = \frac{1}{1 + 10^{(R_B - R_A)/400}}$$

where R_A and R_B are the ratings of Players A and B respectively prior to the tournament. The sum total of E_A for each match played by a player will give their expected score throughout the entire tournament. The byes and forfeited rounds are not considered since there is no actual opponent rating that can be compared against.

```
players_df <- players_df |>
  mutate(
    expected_score = 7 * (1 / (1 + 10^((avg_opponent_pre_rating - pre_rating) /
400))),
    expected_score = round(expected_score, 2),
    score_diff = round(points - expected_score, 2)
  )

head(players_df)
```

pair	name	state	points	pre_rating	avg_opponent_pre_rating	
1	1	GARY HUA	ON	6.0	1794	1605
2	2	DAKSHESH DARURI	MI	6.0	1553	1469
3	3	ADITYA BAJAJ	MI	6.0	1384	1564
4	4	PATRICK H SCHILLING	MI	5.5	1716	1574
5	5	HANSHI ZUO	MI	5.5	1655	1501
6	6	HANSEN SONG	OH	5.0	1686	1519
	expected_score	score_diff				
1	5.24	0.76				
2	4.33	1.67				
3	1.83	4.17				

4	4.86	0.64
5	4.96	0.54
6	5.06	-0.06

```
# Five biggest overperformers
players_df |>
  arrange(desc(score_diff)) |>
  select(name, points, expected_score, score_diff) |>
  head(5)
```

	name	points	expected_score	score_diff
1	ADITYA BAJAJ	6.0	1.83	4.17
2	ANVIT RAO	5.0	1.76	3.24
3	ZACHARY JAMES HOUGHTON	4.5	1.26	3.24
4	JACOB ALEXANDER LAVALLEY	3.0	0.02	2.98
5	AMIYATOSH PWNANANDAM	3.5	0.62	2.88

```
# Five biggest underperformers
players_df |>
  arrange(score_diff) |>
  select(name, points, expected_score, score_diff) |>
  head(5)
```

	name	points	expected_score	score_diff
1	ASHWIN BALAJI	1.0	6.15	-5.15
2	LOREN SCHWIEBERT	3.5	6.30	-2.80
3	GEORGE AVERY JONES	3.5	6.29	-2.79
4	GAURAV GIDWANI	3.5	6.09	-2.59
5	CHIEDOZIE OKORIE	3.5	5.88	-2.38

Conclusion

The overachievers were typically less rated players with scores significantly higher than those expected due to their opponents' rating, while the underachievers were more rated players, including the best rated player rated at 1745, with a score significantly lower than expected. This simply means that the expected score is just an estimation from pre-tournament ratings, hence a significant deviation implies poor performance with respect to their ratings.

In order to further analyze or validate this, one would run a similar comparison with the use of post-tournament ratings in order to determine whether the biggest over/underachievers had also the greatest rating changes, which would be an important internal consistency check. In addition,

one would analyze how sensitive the over/underachievers ranking is to the specific ELO formula being used.